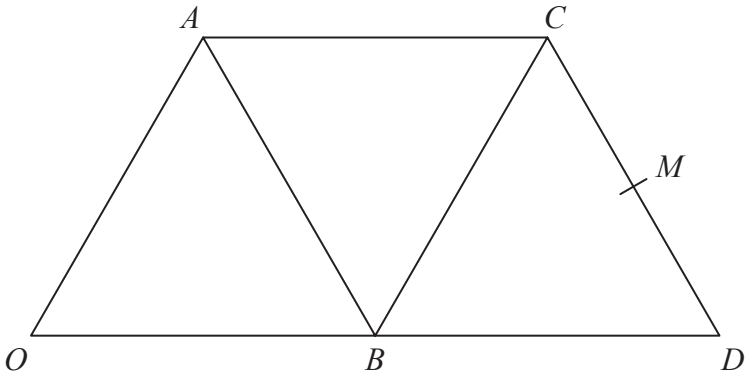
20



$OACD$ is a trapezium and $OACB$ is a parallelogram.
B is the midpoint of OD.
M is the midpoint of CD.

$\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$

Given that $\overrightarrow{BM} = k \times \overrightarrow{OC}$ where k is a scalar,

use a vector method to find the value of k.

Using directed vectors: OC = a + b; OD = 2b.

CD = OD − OC = 2b − (a + b) = b − a.

1 mark
Find CD; halve

M is the midpoint, so CM = ½(b − a).

BM = BC + CM = a + ½(b − a)

1 mark
Complete BM route

= ½(a + b) = ½OC.

Therefore k = ½.

1 mark
k=1/2

(Total for Question 20 is 3 marks)